Bashundhara SMART SPORTS CITY OPERATING SYSTEM

(Bashundhara-SSCOS)

Enterprise User Requirement Specification (URS) & Software Requirement Specification (SRS)

Document Classification: Confidential

Prepared For: Bashundhara Sports City

Revision: v2.0 — updated with expanded testing/security acceptance criteria and new Food, Beverage & Retail Ordering module

Document Control

Distribution List

- Bashundhara Sports City — Project Steering Committee
- Anvir Bashundhara Group — Project ABG-W Team
- Shortlisted Vendors / System Integrators (RFP recipients)
- IT, Security, Legal & Procurement Departments

Revision History

Version	Date	Summary of Change
v1.0	—	Original Enterprise URS/SRS issued for RFP.
v2.0	Current	Added critical-defect (P1) testing/security acceptance requirements (Sections 31, 33); added new Section 22A — Smart Food, Beverage & Retail Ordering, with supporting updates to Sections 4, 6, 7, 11, 13, 17, 22, 23, 24, 26, 27, 28, 29, 30, 31 and 33. All new IDs verified against the existing ID register to avoid duplication (see Appendix A).

Purpose of This Document

This document expands the original ABG-SSCOS vision statement into a vendor-ready Enterprise User Requirement Specification (URS) and Software Requirement Specification (SRS). It is intended to be issued as part of a Request for Proposal (RFP) so that prospective system integrators and technology vendors can size, design, cost, and commit contractually to the delivery of the Bashundhara Smart Sports City Operating System.

Scope of This Document

This document covers: business and technical context, the target system architecture, module-by-module functional requirements, user stories, formal use cases, business process flows, screen/UI requirements, API and integration requirements, data architecture, key performance indicators, acceptance criteria, service level agreements, and cybersecurity requirements. It also defines the vendor evaluation approach to be used during procurement.

How Vendors Should Use This Document

- Respond to every functional requirement (FR) with a compliance status: Comply / Comply with Configuration / Comply with Customization / Not Compliant / Roadmap.
- Reference requirement IDs (e.g., FR-TCK-004, UC-SEC-02) directly in the technical and commercial proposal.
- Provide a mapping of proposed product modules against each layer of the reference architecture in Section 3.
- Complete the acceptance criteria and SLA sections as binding commitments in the contract schedule.

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1. Vision & Business Context

1.1 Vision Statement

ABG-SSCOS shall be a fully integrated, Artificial-Intelligence-powered Smart Sports City Operating System capable of operating, monitoring, automating, securing, analysing, and optimizing every aspect of Bashundhara Sports City from a single unified platform, delivering

"One System, One Dashboard, One Source of Truth"

The platform shall simultaneously function as an ERP, Command Center, Security Platform, Stadium Management Platform, Facility Management Platform, Customer Experience Platform, Digital Twin Platform, AI Decision Engine, Revenue Management Platform, and Enterprise Data Platform.

1.2 Business Objectives

Objective	Description
Unified Operations	Eliminate departmental silos by operating stadium, retail, hospitality, security, and facility functions from one platform.
Revenue Growth	Increase ticketing, membership, retail, and sponsorship revenue through pricing, upsell, and analytics.
Visitor Experience	Deliver frictionless, contactless, personalized experiences across booking, entry, and on-site services.
Safety & Security	Provide proactive, AI-assisted threat detection and incident response across the entire city's footprint.
Operational Efficiency	Reduce manual effort and downtime through predictive maintenance and automated workflows.
Data-Driven Governance	Give executives real-time, trustworthy KPIs for evidence-based decision-making.

1.3 Stakeholders

Stakeholder Group	Representative Roles	Primary Interest
Executive Sponsors	Bashundhara Sports City Board, ABG-W Project Leadership	ROI, strategic control, brand reputation
Operations	Facility Managers, Venue Managers, Event Managers	Day-to-day system usability and reliability
Security	SOC Officers, Security Managers, Emergency Response Teams	Real-time detection, response SLAs

Stakeholder Group	Representative Roles	Primary Interest
Finance & Commercial	Finance Controller, Revenue Manager, Vendor Partners	Accurate billing, settlement, forecasting
HR	HR Manager, Payroll Officer	Workforce management and compliance
Visitors & Members	Public visitors, season ticket holders, corporate guests	Convenience, safety, personalization
IT & Cybersecurity	CTO, IT Infrastructure Team, CISO	System integrity, uptime, compliance
Regulators & Auditors	Government bodies, certification auditors	Legal, safety, and data-protection compliance

1.4 Guiding Design Principles

- Modular, API-first architecture that allows phased rollout and future extensibility.
- Single source of truth: no duplicate master data across modules.
- Security and privacy by design, aligned to ISO 27001 / ISO 27701 / SOC 2 / OWASP.
- AI-assisted, human-supervised automation — no fully autonomous action on life-safety events without human confirmation, except where explicitly approved SOPs allow it.
- Cloud-ready, on-premises-capable hybrid deployment to satisfy data residency requirements.
- Vendor-agnostic integration layer to avoid lock-in on IoT, CCTV, and access-control hardware.

2. Scope

2.1 In Scope

- Design, development/configuration, integration, testing, deployment, and warranty support of ABG-SSCOS across all layers described in Section 3.
- Integration with third-party hardware and software systems listed in Section 28 (Integration Architecture).
- Data migration of existing visitor, membership, and asset records (where applicable).
- Training, documentation, and knowledge transfer to ABG operational staff.

2.2 Out of Scope (unless separately contracted)

- Supply and installation of physical CCTV cameras, turnstiles, and IoT sensors (integration only, unless the vendor is also the hardware supplier).
- Civil, electrical, and mechanical works for the Building Management System field devices.
- Content creation for marketing, sponsorship media, or event production.

3. Reference System Architecture

The platform shall be structured into six architectural layers. Each layer is independently scalable but exposes and consumes standardized APIs through the integration bus described in Section 28.

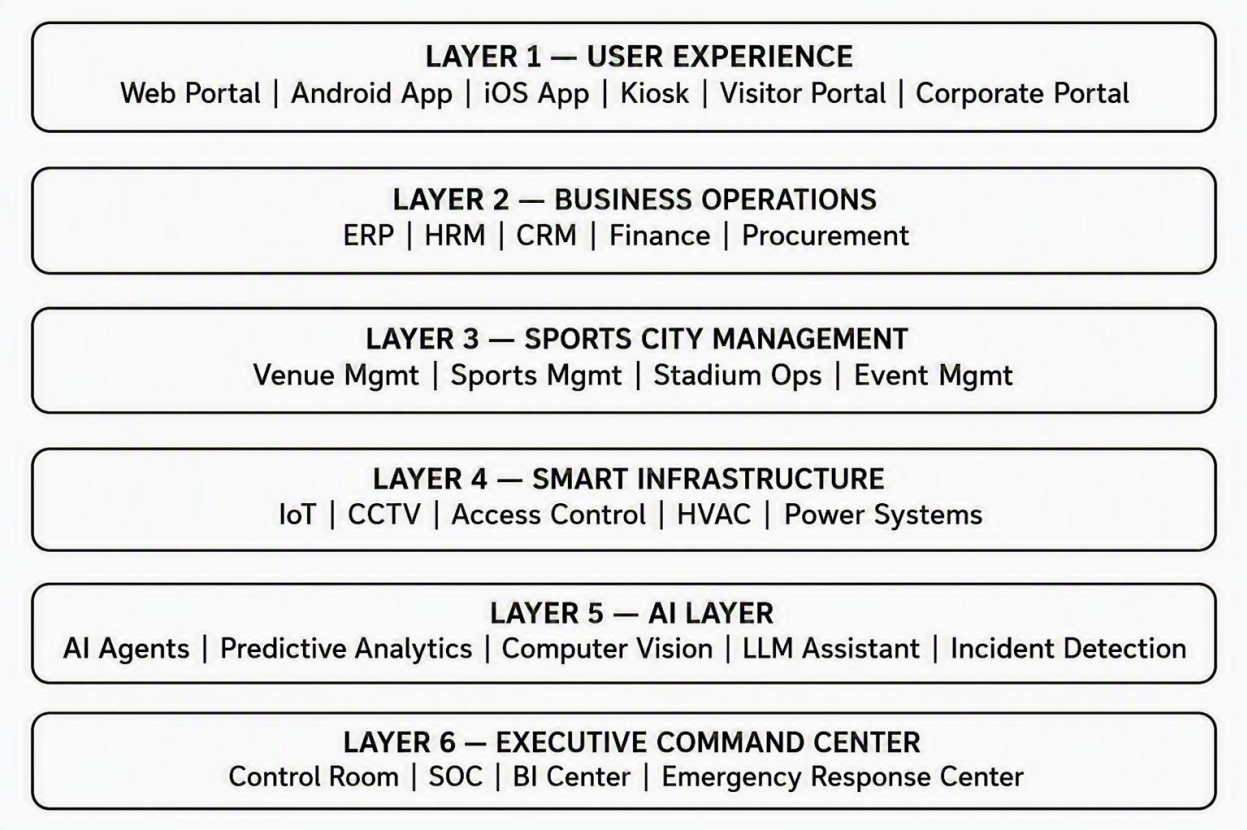


Figure 3.1 — ABG-SSCOS Six-Layer Reference Architecture

Layer	Components	Primary Responsibility
Layer 1: User Experience	Web Portal, Android App, iOS App, Self-Service Kiosk, Visitor Portal, Corporate Portal	All human-facing interaction channels
Layer 2: Business Operations	ERP, HRM, CRM, Finance, Procurement	Core enterprise transaction processing
Layer 3: Sports City Management	Venue Management, Sports Management, Stadium Operations, Event Management	Domain-specific operational management
Layer 4: Smart Infrastructure	IoT, CCTV, Access Control, HVAC, Power Systems	Physical asset connectivity and telemetry

Layer	Components	Primary Responsibility
Layer 5: AI Layer	AI Agents, Predictive Analytics, Computer Vision, LLM Assistant, Incident Detection	Intelligence, automation, and decisioning
Layer 6: Executive Command Center	Real-Time Control Room, SOC, BI Center, Emergency Response Center	Cross-domain situational awareness and control

3.1 Non-Functional Architecture Requirements

Attribute	Requirement
Availability	99.9% uptime for Layer 1/2/3 transactional systems; 99.95% for Layer 4/6 life-safety and security functions (see SLA Section 32).
Scalability	Support horizontal scaling to a minimum of 100,000 registered visitor profiles and 20,000 concurrent app/portal sessions during peak events.
Latency	Access-control decision (face/QR/RFID scan to turnstile release) shall complete in ≤ 1.5 seconds at the 95th percentile.
Interoperability	All modules shall expose RESTful JSON APIs and support webhook-based event publishing (see Section 27).
Deployment Model	Hybrid: Cloud-hosted application tier with on-premises edge nodes for CCTV/IoT processing and local failover.
Data Residency	Personally identifiable information (PII) of Bangladeshi citizens shall be stored in-country unless otherwise approved by ABG legal/compliance.
Disaster Recovery	RPO ≤ 15 minutes, RTO ≤ 4 hours for core transactional systems; RPO ≤ 1 hour, RTO ≤ 1 hour for SOC/life-safety systems.

FUNCTIONAL REQUIREMENTS

Requirement priority follows the MoSCoW method: Must (mandatory for go-live), Should (high value, next release acceptable), Could (desirable, roadmap acceptable). Vendors shall state compliance against every ID individually.

4. Functional Requirements — e-Ticketing & Smart Access

The ticketing platform is the primary access-control mechanism for Bashundhara Sports City and shall govern all visitors, members, and staff entry.

ID	Requirement	Priority
FR-TCK-001	The system shall allow visitor self-registration via Web, App, and Kiosk capturing Name, NID/Passport, Mobile, Email, Date of Birth.	Must
FR-TCK-002	The system shall support face registration and optional biometric profile capture at the point of ticket issuance.	Must
FR-TCK-003	The system shall support sale of Sports Tickets (Football, Futsal, Padel, Cricket, Tennis, Swimming, Gym, Academy).	Must
FR-TCK-004	The system shall support free passes for Visitor (Tourist Visit, Gallery Visit, Restaurant Access, Shopping Access).	Must
FR-TCK-005	The system shall support Premium Ticket categories (VIP, VVIP, Corporate, Media) with distinct pricing and access rules.	Must
FR-TCK-006	The system shall support Membership types (Annual, Family, Student, Corporate) including renewal and auto-billing.	Must
FR-TCK-007	The system shall generate a unique QR/RFID/NFC credential per ticket or membership, valid for a configurable time window.	Must
FR-TCK-008	The system shall allow refunds, cancellations, and ticket transfers subject to configurable business rules.	Should
FR-TCK-009	The system shall support dynamic and promotional pricing rules (early-bird, off-peak, bundle) configurable by admin.	Should
FR-TCK-010	The system shall reconcile ticketing revenue with Finance ERP General Ledger on a daily automated batch.	Must
FR-TCK-011 (NEW)	In addition to the QR/RFID/NFC credential in FR-TCK-007, the system shall support an optional wearable wristband/tag as a physical credential and cashless payment instrument, linked to the same visitor wallet used for ticketing and in-facility purchases (see FR-FNB-005). The wristband shall operate via the RFID/NFC modalities already defined for turnstile authentication in FR-ACC-005 — no separate access-control mechanism is introduced.	Should

FR-TCK-011 is new in v2.0. It extends the existing credential/payment model rather than duplicating FR-ACC-005 (multi-modal turnstile authentication), which already supports RFID/NFC as a scan method.

5. Functional Requirements — AI-Based Access Control

Access decisions across all restricted zones shall be automated and auditable.

ID	Requirement	Priority
FR-ACC-001	The system shall perform real-time face recognition against the enrolled visitor/employee database at all configured access points.	Must
FR-ACC-002	The system shall check every access attempt against a blacklist database and deny/flag matches within 1 second.	Must
FR-ACC-003	The system shall recognize VIP/VVIP profiles and trigger a configurable concierge notification workflow.	Should
FR-ACC-004	The system shall recognize repeat visitors and surface visit-history context to front-desk/concierge staff.	Could
FR-ACC-005	The system shall support multi-modal turnstile authentication: Face Scan, QR Scan, RFID, NFC, selectable per gate.	Must
FR-ACC-006	The system shall enforce role-based restricted-area access for Players, Coaches, Employees, Vendors, Visitors, VIP Guests based on configurable zone permissions.	Must
FR-ACC-007	The system shall log every access grant/denial event with timestamp, identity, zone, and method for audit purposes for a minimum of 12 months.	Must
FR-ACC-008	The system shall allow security operators to manually override an automated access decision, with mandatory reason capture.	Must

6. Functional Requirements — Smart Security Operations Center (SOC)

The SOC module shall aggregate all security-relevant sensors and video feeds into unified, AI-assisted monitoring.

ID	Requirement	Priority
FR-SOC-001	The system shall display live CCTV feeds from all connected cameras on a unified video wall interface with configurable layouts.	Must
FR-SOC-002	The system shall run video analytics for facial recognition, intruder detection, suspicious object detection, and violence detection.	Must
FR-SOC-003	The system shall provide AI-based crowd density estimation and generate alerts when configurable thresholds are breached.	Must
FR-SOC-004	The system shall detect fire and smoke through video analytics and/or integrated fire-alarm sensor feeds.	Must
FR-SOC-005	The system shall flag unauthorized access attempts in real time to SOC operators with location and camera reference.	Must
FR-SOC-006	The system shall provide an emergency call management console linking incoming emergency calls to nearest camera/incident context.	Should
FR-SOC-007	The system shall retain video analytics metadata (not necessarily raw footage) for a minimum of 90 days for investigation.	Must
FR-SOC-008 (NEW)	The system shall provide a Lost & Found register allowing staff to log a found item with its location, timestamp, and linked CCTV reference, and allow visitors to search/report a lost item via the app or help desk.	Should
FR-SOC-009 (NEW)	The system shall support an optional child-safety mode in which a minor's wristband/tag (see FR-TCK-011) is paired to a guardian's profile; the system shall alert staff and the guardian if the pair separates beyond a configurable zone boundary.	Could

7. Functional Requirements — AI Incident Response Management

Incident detection, response, and closure shall follow a consistent, trackable workflow across incident types.

ID	Requirement	Priority
FR-INC-001 (UPDATED)	The system shall automatically detect and classify incidents: Fire, Medical Emergency, Crowd Stampede, Security Breach, Vehicle Collision, and Severe Weather Event (see FR-INC-008).	Must
FR-INC-002	The system shall automatically trigger the configured SOP workflow for each incident classification.	Must
FR-INC-003	The system shall dispatch alerts to the relevant team (Security, Fire, Medical) via app push, SMS, and/or radio integration.	Must
FR-INC-004	The system shall send stakeholder notifications (management, PR, safety officer) based on incident severity tier.	Should
FR-INC-005	The system shall track response time (detection to first responder dispatch) and resolution time (dispatch to closure).	Must
FR-INC-006	The system shall require a closure report with root cause, actions taken, and responsible officer before an incident can be marked closed.	Must
FR-INC-007	The system shall provide a post-incident analytics dashboard showing incident trends by type, location, and time.	Should
FR-INC-008 (NEW)	The system shall ingest a live weather data feed (see Section 28.1) and automatically trigger the Severe Weather SOP for outdoor facilities (football, cricket, swimming, tennis, padel) when lightning, storm, or extreme-heat thresholds configured by ABG are breached.	Should

8. Functional Requirements — Command & Control Center

A unified 24/7 dashboard suite for executive and operational decision-making.

ID	Requirement	Priority
FR-CCC-001	The system shall provide a Security Dashboard showing live CCTV thumbnails, active alerts, and real-time visitor flow heatmaps.	Must
FR-CCC-002	The system shall provide a Facility Dashboard showing live electricity, water, and HVAC consumption and status.	Must
FR-CCC-003	The system shall provide a Revenue Dashboard showing ticket sales, venue revenue, and retail revenue in real time.	Must
FR-CCC-004	The system shall provide an Executive Dashboard showing occupancy, revenue, open incidents, and configurable KPIs.	Must
FR-CCC-005	All dashboards shall support role-based visibility and drill-down from summary KPI to transaction-level detail.	Must
FR-CCC-006	Dashboards shall be accessible on large-format video-wall displays as well as desktop and tablet browsers.	Should

9. Functional Requirements — Sports Operations Management

Covers booking, tournaments, and academy operations across all sports facilities.

ID	Requirement	Priority
FR-SPT-001	The system shall support court, ground, and slot booking with real-time availability calendars.	Must
FR-SPT-002	The system shall support tournament fixture creation, referee assignment, team registration, and match scheduling.	Must
FR-SPT-003	The system shall support academy student enrollment, coach assignment, and attendance tracking.	Must
FR-SPT-004	The system shall track athlete performance data, health data (subject to consent and data-protection controls), equipment usage, and training plans.	Should
FR-SPT-005	The system shall send automated reminders for upcoming bookings, tournaments, and academy sessions.	Should
FR-SPT-006	The system shall prevent double-booking of any court, ground, or slot resource.	Must

10. Functional Requirements — Smart Facility Management

Covers maintenance and operational SOPs for stadiums, courts, pools, gyms, accommodation, and restaurants.

ID	Requirement	Priority
FR-FAC-001	The system shall maintain a maintenance schedule per asset (stadium, courts, pools, gym, accommodation, restaurants).	Must
FR-FAC-002	The system shall support SOP-driven cleaning schedules and daily inspection checklists with photo evidence capture.	Must
FR-FAC-003	The system shall support work-order creation, assignment, tracking, and closure with SLA timers.	Must
FR-FAC-004	The system shall automatically escalate overdue maintenance/cleaning tasks to the next management tier.	Must
FR-FAC-005	Example workflow: for swimming pool cleaning, the system shall create the schedule, assign staff, verify completion (via checklist/photo), and issue escalation if overdue.	Must

11. Functional Requirements — Building Management System (BMS)

Integration with HVAC, generator, UPS, lift, escalator, water pump, and lighting systems.

ID	Requirement	Priority
FR-BMS-001	The system shall provide live monitoring of HVAC, Generator, UPS, Lift, Escalator, Water Pump, and Lighting systems.	Must
FR-BMS-002	The system shall allow authorized operators to remotely control supported BMS equipment (start/stop/setpoint change).	Should
FR-BMS-003	The system shall provide energy optimization recommendations based on occupancy and usage patterns.	Should
FR-BMS-004	The system shall provide predictive maintenance alerts for BMS equipment based on runtime hours and sensor trends.	Should
FR-BMS-005 (NEW)	Building on the consumption data used for the energy-optimization recommendations in FR-BMS-003, the system shall consolidate energy, water, and waste data into a Sustainability/ESG reporting dashboard (see Section 26.4) with configurable reduction targets and period-over-period trend reporting suitable for external ESG disclosure.	Should

12. Functional Requirements — Power Station Management

Covers substation, transformer, and generator monitoring across the site.

ID	Requirement	Priority
FR-PWR-001	The system shall monitor substation and transformer health parameters (load, temperature, voltage) in real time.	Must
FR-PWR-002	The system shall monitor generator status and fuel consumption with low-fuel alerting.	Must
FR-PWR-003	The system shall provide load-balancing recommendations across available power sources.	Should
FR-PWR-004	The system shall provide predictive failure analysis for electrical infrastructure based on historical trend data.	Should

13. Functional Requirements — Parking & Traffic Management

AI-based vehicle recognition, allocation, and congestion analytics.

ID	Requirement	Priority
FR-PRK-001	The system shall perform ANPR-based vehicle recognition at all parking entry/exit points.	Must
FR-PRK-002	The system shall provide smart parking allocation and real-time slot availability to visitors via app/kiosk.	Must
FR-PRK-003	The system shall support advance parking reservations tied to event or ticket bookings.	Should
FR-PRK-004	The system shall manage parking revenue including tariff configuration and automated billing.	Must
FR-PRK-005	The system shall provide traffic congestion analytics and suggest alternate routing during peak events.	Could
FR-PRK-006 (NEW)	The system shall integrate with third-party ride-hailing platforms (e.g., Pathao, Uber) to offer a coordinated pickup point and suggested pickup time aligned to event end-times, to reduce egress congestion.	Could

14. Functional Requirements — Employee Management Suite

HRM, payroll, and workforce-tracking functions for all Bashundhara Sports City staff.

ID	Requirement	Priority
FR-HRM-001	The system shall support recruitment, onboarding, and full employee lifecycle management.	Must
FR-HRM-002	The system shall support payroll processing including statutory deductions applicable in Bangladesh.	Must
FR-HRM-003	The system shall support leave management with configurable approval workflows.	Must
FR-HRM-004	The system shall support performance management including goal-setting and periodic review cycles.	Should
FR-HRM-005	The system shall capture attendance via Face Recognition, GPS, and Biometric devices.	Must
FR-HRM-006	The system shall track field employee location, assigned duty, and task progress in real time where applicable.	Should

15. Functional Requirements — Vendor & Business Partner Management

Supports revenue-share commercial models with food courts, shops, and brand partners.

ID	Requirement	Priority
FR-VND-001	The system shall track vendor sales/revenue for food courts, shops, sports stores, and brand partners.	Must
FR-VND-002	The system shall support automated revenue-share settlement calculation and reporting.	Must
FR-VND-003	The system shall maintain vendor contract records including terms, renewal dates, and commission rates.	Must
FR-VND-004	The system shall provide vendor performance analytics (footfall conversion, average sale, category trends, etc).	Should

16. Functional Requirements — Event & Sponsor Management

Covers event bookings, sponsorships, and ROI tracking.

ID	Requirement	Priority
FR-EVT-001	The system shall support event booking including venue, date/time, resource, and vendor coordination.	Must
FR-EVT-002	The system shall manage sponsorship packages, inventory, and allocation to sponsors.	Must
FR-EVT-003	The system shall track ticket allocation tied to specific events and sponsor entitlements.	Must
FR-EVT-004	The system shall provide brand-visibility and sponsor ROI analytics post-event.	Should

17. Functional Requirements — Customer Experience Platform (AI Concierge)

Conversational and self-service support across digital channels.

ID	Requirement	Priority
FR-CXP-001	The system shall provide an AI concierge capable of ticket support, facility guidance, event information, and venue navigation.	Must
FR-CXP-002	The AI concierge shall be available via App, Website, WhatsApp, Messenger, and Voice Bot.	Should
FR-CXP-003	The AI concierge shall escalate unresolved queries to a human agent with full conversation context.	Must
FR-CXP-004	The AI concierge shall support Bengali and English at minimum.	Must
FR-CXP-005 (NEW)	In addition to the Bengali/English minimum in FR-CXP-004, the AI concierge shall support additional languages as configurable add-ons for major international events (e.g., international tournaments, corporate/sponsor delegations).	Could

18. Functional Requirements — Hospitality & Accommodation Management

Applicable where guest housing exists on site.

ID	Requirement	Priority
FR-HSP-001	The system shall support room reservation, check-in, and check-out workflows.	Should
FR-HSP-002	The system shall generate consolidated guest billing across room, F&B, and facility usage.	Should
FR-HSP-003	The system shall manage housekeeping task assignment and completion tracking.	Should

19. Functional Requirements — Digital Twin & Virtual Tour

A 3D virtual replica of Bashundhara Sports City for navigation, planning, and simulation.

ID	Requirement	Priority
FR-DGT-001	The system shall maintain an interactive 3D map of the Sports City accessible via web/app.	Should
FR-DGT-002	The system shall provide visitor route planning within the 3D map.	Could
FR-DGT-003	The system shall support emergency evacuation simulation for staff training and planning.	Should
FR-DGT-004	The system shall support event-layout simulation prior to live events.	Could

20. Functional Requirements — Finance & Accounting ERP

Core financial modules with AI-assisted controls.

ID	Requirement	Priority
FR-FIN-001	The system shall provide General Ledger, Accounts Payable, Accounts Receivable, Budgeting, Cost Centers, and Fixed Assets modules.	Must
FR-FIN-002	The system shall support bank reconciliation and VAT/taxation calculation per Bangladesh NBR requirements.	Must
FR-FIN-003	The system shall provide AI-based auto voucher suggestion from source transactions.	Should
FR-FIN-004	The system shall provide AI-based fraud and anomaly detection on financial transactions.	Should
FR-FIN-005	The system shall provide AI-assisted revenue forecasting by revenue stream.	Should

21. Functional Requirements — Procurement & Inventory

End-to-end procure-to-pay and stock management.

ID	Requirement	Priority
FR-PRC-001	The system shall support purchase requisition, RFQ, vendor selection, and purchase order issuance.	Must
FR-PRC-002	The system shall support Goods Receipt Note (GRN) and three-way match against PO and invoice.	Must
FR-PRC-003	The system shall maintain stock management with reorder-point alerts.	Must
FR-PRC-004	The system shall track equipment lifecycle from procurement to disposal.	Should

22. Functional Requirements — Business Intelligence & AI Copilot

Predictive analytics and natural-language query capability for operational and executive users.

ID	Requirement	Priority
FR-BI-001	The system shall provide footfall, revenue, event-demand, and utility-consumption forecasts.	Must
FR-BI-002	The system shall provide an AI Copilot capable of answering natural-language questions such as “How much revenue did we earn today?”, “How many visitors are currently inside?”, and “Which facilities need maintenance?” with sourced, auditable answers.	Must
FR-BI-003	The AI Copilot shall restrict its answers to data the requesting user's role is authorized to view.	Must
FR-BI-004 (NEW)	Building on the footfall forecasting in FR-BI-001, the system shall provide real-time predicted wait-times at concessions, restrooms, and access gates, surfaced on the visitor app, kiosks, and Command Center dashboards (Section 26.4).	Should

22A. Functional Requirements — Smart Food, Beverage & Retail Ordering (NEW)

This is a new module added in v2.0. Physical access to restaurants and shops is already governed by the free-pass rules in Section 4 (FR-TCK-004); this section defines the in-facility ordering, fulfillment, and payment experience once a visitor is inside the venue — for example, ordering food from any restaurant or food court in the facility from wherever the visitor currently is, with the system determining the visitor's location via its own indoor map and routing the order accordingly.

ID	Requirement	Priority
FR-FNB-001	The system shall display all active restaurants and food-court outlets available for ordering, filtered and ranked by the visitor's current in-venue location.	Must
FR-FNB-002	The system shall determine the visitor's in-venue location using the platform's own indoor positioning (Wi-Fi/BLE beacon or Digital Twin zone mapping per FR-DGT-001), rather than device GPS alone, given the accuracy needed indoors.	Must
FR-FNB-003	The system shall route each order to the correct outlet's kitchen/POS queue and provide the visitor a live order-status timeline (received → preparing → ready → delivered/picked up).	Must
FR-FNB-004	The system shall support two fulfillment modes, configurable per outlet: pickup (with a designated collection point/QR) and seat- or location-delivery (via staff or delivery-robot integration, where available).	Must
FR-FNB-005	The system shall process payment through the same wallet/payment gateway used for ticketing (Section 4) and the optional wearable credential (FR-TCK-011), with a single consolidated visitor transaction history.	Must
FR-FNB-006	The system shall reconcile food and retail order revenue per outlet automatically with the Vendor Revenue-Share module (FR-VND-001, FR-VND-002); order-level sales shall not be recorded a second time outside that module.	Must
FR-FNB-007	The system shall allow outlets to mark menu/catalog items out-of-stock or set prep-time estimates in real time, reflected instantly to the visitor.	Should
FR-FNB-008	The system shall notify the visitor and offer reassignment or refund if an outlet cannot fulfill an order within a configurable SLA window.	Should

ID	Requirement	Priority
FR-FNB-009	The AI Concierge (FR-CXP-001) shall be able to take a food or retail order conversationally as an alternative to browsing the outlet list.	Could
FR-FNB-010	The system shall extend the same ordering experience to merchandise and retail shops (sports stores, brand partners) already tracked under FR-VND-001, allowing browsing and purchase of retail items through the app/kiosk.	Should
FR-FNB-011	The system shall support in-venue pickup-counter or locker fulfillment for merchandise orders placed under FR-FNB-010.	Should

23. User Stories

User stories are expressed in the standard “As a … I want … so that …” format and are traceable to the functional requirement IDs in the preceding sections. Vendors should use these to validate that proposed configurations meet real operational intent, not just literal requirement text.

ID	Persona	I want…	So that…	Traceability
US-01	Visitor	book a swimming pool slot from the mobile app and pay online	secure my preferred time without visiting the counter in person	FR-TCK-001, FR-TCK-007, FR-SPT-001
US-02	Visitor	enter the venue by simply scanning my face at the gate	not carry a physical ticket or card	FR-TCK-002, FR-ACC-001, FR-ACC-005
US-03	Security Operator	see an automatic alert with camera reference when an intruder is detected in a restricted zone	dispatch a response team immediately	FR-SOC-002, FR-SOC-005, FR-INC-003
US-04	SOC Manager	see crowd density heatmaps in real time during a major match	proactively prevent stampede risk	FR-SOC-003, FR-CCC-001
US-05	Facility Manager	have the system auto-escalate an overdue pool-cleaning task to my supervisor	hygiene SOPs are never silently missed	FR-FAC-002, FR-FAC-004, FR-FAC-005
US-06	Academy Coordinator	enroll a new academy student and assign a coach in one workflow	onboarding is fast and error-free	FR-SPT-003
US-07	Finance Controller	have daily ticketing revenue reconcile automatically with the General Ledger	not need manual journal entries every day	FR-TCK-010, FR-FIN-001
US-08	Vendor Partner (Food Court)	see my daily sales and revenue-share settlement in a partner portal	trust the commission calculation without disputes	FR-VND-001, FR-VND-002
US-09	Executive	see a single dashboard showing today's revenue,	make decisions without calling five different departments	FR-CCC-004

ID	Persona	I want…	So that…	Traceability
		current occupancy, and open incidents		
US-10	Executive	ask “Which facilities need maintenance?” in plain language and get an instant answer	not navigate multiple reports	FR-BI-002
US-11	HR Manager	have field staff attendance captured via GPS and face recognition automatically	payroll input requires no manual timesheets	FR-HRM-005, FR-HRM-002
US-12	Parking Attendant	have the system recognize a vehicle's plate and allocate the nearest available slot	traffic doesn't back up at the gate during events	FR-PRK-001, FR-PRK-002
US-13	Event Manager	allocate sponsor tickets automatically against the sponsorship package inventory	never oversell sponsor entitlements	FR-EVT-002, FR-EVT-003
US-14	Visitor (Emergency)	reach a real person via WhatsApp concierge and have them see my whole booking history	not repeat my problem	FR-CXP-001, FR-CXP-003
US-15	Emergency Response Officer	have the system automatically alert the medical team and log the response time when a medical emergency is detected	response performance can be measured and improved	FR-INC-001, FR-INC-003, FR-INC-005
US-16	IT Administrator	have role-based access enforced on every dashboard and API	sensitive financial and security data is never exposed to unauthorized users	FR-CCC-005, FR-BI-003, Section 33
US-17 (NEW)	Visitor	order food from any restaurant in the facility without leaving my seat or current location	I don't lose time queuing during a match or event	FR-FNB-001, FR-FNB-002, FR-FNB-003, FR-FNB-005

24. Use Cases

The following formal use cases illustrate end-to-end system behavior for the highest-risk and highest-value scenarios. Vendors shall confirm their proposed solution can execute each flow, including all listed alternate/exception paths, during the proof-of-concept stage.

UC-TCK-01 — Purchase a Sports Ticket and Gain Venue Access

Field	Detail
Actors	Visitor (Primary); Payment Gateway, Access Control System (Supporting)
Trigger	Visitors want to play a sport (e.g., Padel) at Bashundhara Sports City.
Preconditions	Visitor has a registered account or completes registration during booking.

Main Flow:

- 1. Visitor opens the App/Web Portal and selects the desired sport and slot.
- 2. System displays available slots and pricing.
- 3. Visitor selects a slot and proceeds to payment.
- 4. System processes payment through the integrated payment gateway.
- 5. System issues a digital ticket with QR/RFID credential and sends confirmation via SMS/Email.
- 6. On the day of visit, visitor presents the QR code or face at the turnstile.
- 7. System verifies identity/credential, checks blacklist status, and confirms booking validity.
- 8. System grants access and logs entry with timestamp.

Alternate / Exception Flows:

- 3a. Payment fails → system retains slot hold for 10 minutes and prompts retry.
- 7a. Blacklist match found → system denies access and alerts SOC immediately (see UC-SEC-01).
- 7b. Booking expired or slot mismatch → system denies entry and directs visitors to help desk / AI concierge.

Postconditions:

Visitor is inside the venue; entry is logged; revenue is recorded against the correct sport and cost center.

UC-SEC-01 — Detect and Respond to an Unauthorized Access Attempt

Field	Detail
Actors	AI Access Control Engine (Primary); SOC Operator, Security Field Team (Supporting)

Field	Detail
Trigger	An individual attempts to enter a restricted zone without valid credentials, or a blacklisted identity is recognized at any access point.
Preconditions	Face recognition and blacklist databases are active and synced; access point is online.

Main Flow:

- 1. Individual presents at an access point (turnstile, restricted door).
- 2. System scans identity via face/QR/RFID/NFC.
- 3. System checks identity against permission rules and blacklist database.
- 4. System determines the individual is not authorized for the zone.
- 5. System denies physical access and raises a real-time alert to the SOC dashboard with camera feed reference.
- 6. SOC operator reviews the alert and dispatches the nearest security field team.
- 7. Field team resolves the situation and logs the outcome in the incident module.

Alternate / Exception Flows:

- 4a. Individual is a recognized VIP without pre-registered zone access → system prompts SOC for manual override with reason capture.
- 6a. No field team available within SLA → system auto-escalates to duty manager.

Postconditions:

Access denial and resolution are logged; incident record created if escalated; blacklist database updated if new threat identified.

UC-INC-01 — Automated Fire Detection and Emergency Response

Field	Detail
Actors	AI Incident Detection Engine (Primary); Fire Team, Command Center, Visitors (Supporting)
Trigger	Smoke/fire signature detected via CCTV video analytics or integrated fire-alarm sensor.
Preconditions	Fire detection sensors and/or video analytics are operational and integrated.

Main Flow:

- 1. System detects a fire/smoke signature above the confidence threshold.
- 2. System classifies the incident as “Fire” and determines the affected zone.
- 3. System automatically triggers the Fire SOP workflow.

- 4. System alerts the Fire Team, Security, and Command Center simultaneously via app push and SMS.
- 5. System displays the incident on the Executive/Security dashboard with live camera feed and evacuation zone map.
- 6. Field teams respond; system tracks response time from detection to first responder arrival.
- 7. Once resolved, the responsible officer files a closure report with root cause and actions taken.
- 8. System logs resolution time and closes the incident.

Alternate / Exception Flows:

- 1a. False positive suspected → SOC operator can downgrade/cancel the alert with mandatory justification, logged for audit.
- 4a. Primary responder team does not acknowledge within SLA → system auto-escalates to secondary responder and management.

Postconditions:

Incident fully logged with timestamps, responders, and resolution; feeds into post-incident analytics (FR-INC-007).

UC-FAC-01 — Scheduled Facility Maintenance with Escalation

Field	Detail
Actors	Facility Management System (Primary); Maintenance Staff, Facility Manager (Supporting)
Trigger	Scheduled maintenance/cleaning task becomes due (e.g., swimming pool cleaning).
Preconditions	Maintenance schedule and SOP checklist configured for the asset.

Main Flow:

- 1. System auto-generates the task per the configured schedule.
- 2. System assigns the task to the designated staff member and sends a notification.
- 3. Staff member performs the task and submits completion evidence (checklist + photo) via mobile app.
- 4. System verifies completion against the SOP checklist.
- 5. System marks the task complete and updates the asset maintenance history.

Alternate / Exception Flows:

- 3a. Task not completed within the SLA window → system sends a reminder, then escalates to the Facility Manager if still overdue.

- 4a. Submitted evidence fails verification (e.g., incomplete checklist) → system reopens the task and notifies the assigned staff.

Postconditions:

Asset maintenance history updated; overdue-task escalation log available for audit and vendor SLA reporting.

UC-BI-01 — Executive Natural-Language Query via AI Copilot

Field	Detail
Actors	Executive User (Primary); AI Copilot / BI Engine (Supporting)
Trigger	Executive wants an operational answer without navigating multiple reports.
Preconditions	User is authenticated with an executive role profile; underlying data pipelines are current.

Main Flow:

- 1. Executive opens the AI Copilot chat panel on the Executive Dashboard.
- 2. Executive types a natural-language question, e.g., “How much revenue did we earn today?”
- 3. System interprets intent, checks the user's data-access permissions, and queries the curated data warehouse.
- 4. System returns a direct answer with the supporting figure, time range, and a link to the underlying report.
- 5. Executive may ask a follow-up question in the same context (e.g., “Break that down by revenue stream”).

Alternate / Exception Flows:

- 3a. Requested data is outside the user's authorization → system declines and logs the attempt.
- 3b. Underlying data pipeline is delayed → system states the data's last-refreshed time rather than presenting stale data as current.

Postconditions:

Executive receives an accurate, permission-scoped answer; query is logged for AI Copilot quality monitoring.

UC-VND-01 — Automated Vendor Revenue-Share Settlement

Field	Detail
Actors	Finance System (Primary); Vendor Partner, Finance Controller (Supporting)
Trigger	End of settlement period (e.g., daily or monthly, per contract) is reached.

Field	Detail
Preconditions	Vendor contract terms (commission %, settlement frequency) are configured in the system.

Main Flow:

- 1. System aggregates the vendor's recorded sales for the settlement period.
- 2. System applies the contracted commission/revenue-share formula.
- 3. System generates a settlement statement and posts it to the vendor's partner portal.
- 4. System notifies the vendor and Finance Controller of the settlement outcome.
- 5. Finance Controller reviews and approves the payment run.

Alternate / Exception Flows:

- 1a. Vendor disputes a sales figure → vendor raises a dispute in-portal; settlement is held pending Finance review.

Postconditions:

Settlement statement archived; payment instruction generated for Finance/banking integration.

UC-FNB-01 — Order Food & Beverage via Location-Aware Ordering (NEW)

Field	Detail
Actors	Visitor (Primary); Indoor Positioning System, Outlet POS, Payment Gateway (Supporting)
Trigger	Visitor wants to order food or a drink without leaving their seat or current location inside the venue.
Preconditions	Visitor is inside the venue with an active session; at least one outlet near the visitor's zone is open and accepting orders.

Main Flow:

- 1. Visitor opens the Food & Beverage Ordering screen in the app.
- 2. System determines the visitor's current zone via indoor positioning (FR-FNB-002) and lists nearby open outlets (FR-FNB-001).
- 3. Visitor selects an outlet, browses the menu, and adds items to the cart.
- 4. Visitor chooses a fulfillment mode: pickup point or location-delivery (FR-FNB-004).
- 5. Visitor pays via the shared wallet/payment gateway (FR-FNB-005).
- 6. System routes the order to the outlet's POS queue and displays a live status timeline (FR-FNB-003).
- 7. Outlet prepares and marks the order ready; the visitor is notified.

- 8. Order is fulfilled (collected or delivered); revenue reconciles automatically to the vendor's settlement record (FR-FNB-006).

Alternate / Exception Flows:

- 3a. An item goes out of stock while browsing → system reflects the outlet's real-time update (FR-FNB-007) and removes the item from the cart.
- 7a. Outlet cannot fulfill within the SLA window → system offers reassignment to a comparable outlet or an automatic refund (FR-FNB-008).

Postconditions:

Order is logged end-to-end with timestamps and location; revenue is recorded once against the correct outlet, with no duplicate entry in the vendor settlement ledger.

25. Business Process Diagrams

The diagrams below illustrate the three business processes with the greatest operational and safety impact. Additional process diagrams (booking cancellation, work-order lifecycle, vendor settlement, procurement-to-pay, and the new food/retail ordering flow) shall be jointly finalized with the selected vendor during the design phase.

25.1 e-Ticketing & Smart Access Process



Figure 25.1 — End-to-end visitor ticketing and access flow, from booking to logged entry.

25.2 AI Incident Detection & Response Process

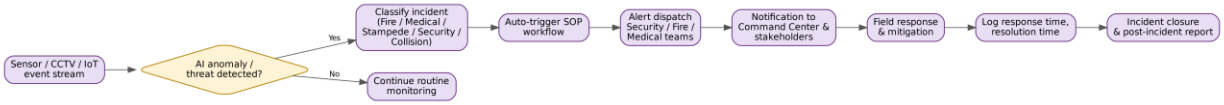


Figure 25.2 — Automated incident classification, SOP trigger, response, and closure tracking.

25.3 Data Flow: Source Systems to Executive Insight

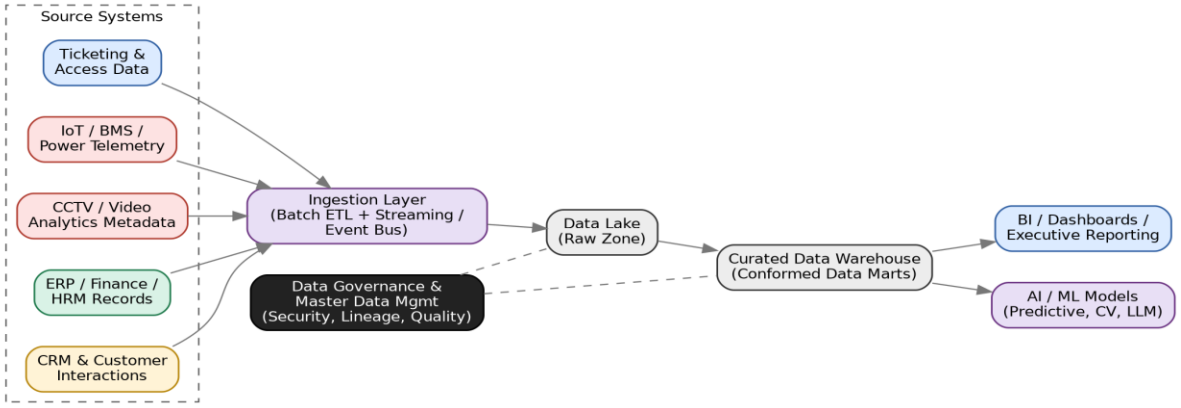


Figure 25.3 — Data ingestion, governance, and consumption path referenced in Section 29 (Data Architecture).

26. Screen / UI Requirements

The screen inventory below defines the minimum set of screens each portal/console shall provide. Vendors shall submit wireframes or existing product screenshots mapped against this inventory as part of the technical proposal. Final UI/UX shall follow the visual identity guidelines to be supplied separately by Bashundhara Sports City.

26.1 Visitor Mobile App (iOS/Android)

Screen	Key Content / Function
Onboarding & Registration	Sign-up, NID/passport capture, face enrolment, consent capture.
Home / Discover	Featured sports, events, promotions; quick-book shortcuts.
Booking & Slot Selection	Calendar view, sport/facility filter, real-time availability, price display.
Payment & Checkout	Payment method selection, promo code entry, order summary, and receipt.
My Tickets / Wallet	QR/RFID credential display, membership card, booking history.
Digital Twin / Navigation	3D map, indoor wayfinding, points of interest.
Food & Beverage / Retail Ordering (NEW)	Nearby outlet list ranked by current location (FR-FNB-001/002), menu/catalog browsing, cart, live order-status tracker, pickup/delivery choice (Section 22A).
AI Concierge Chat	Text/voice chat, quick-action buttons, human hand-off; conversational ordering per FR-FNB-009.
Notifications	Booking reminders, incident/safety alerts, promotions, order-ready alerts.
Profile & Settings	Personal data, payment methods, wearable-credential linkage (FR-TCK-011), and privacy controls.

26.2 Web Portal (Public & Corporate)

Screen	Key Content / Function
Public Landing Page	Facility showcase, event calendar, membership plans.
Booking Engine	Same capability as app, optimized for desktop/kiosk-scale screens.

Screen	Key Content / Function
Corporate Portal	Corporate membership management, bulk ticket allocation, invoicing.
Sponsor Portal	Sponsorship package status, brand-visibility reports, ROI dashboard.
Vendor / Partner Portal	Sales dashboard, settlement statements (incl. food/retail orders per FR-FNB-006), contract documents, dispute raising.

26.3 Self-Service Kiosk

Screen	Key Content / Function
Walk-in Ticket Purchase	Simplified touch flow: sport/event select → pay → print/QR issue.
Membership Check-in	Face/QR scan for existing members and season-ticket holders.
Wayfinding	Static map with “you are here” indicator and directions to key facilities.

26.4 Command & Control Center (Operator Consoles)

Screen	Key Content / Function
Security Dashboard	Live CCTV grid, active alert feed, visitor-flow heatmap, gate status.
Facility Dashboard	Electricity/water/HVAC live gauges, active work orders, BMS equipment status.
Revenue Dashboard	Ticket sales, venue revenue, retail revenue (incl. F&B/retail orders), trend charts, target vs actual.
Executive Dashboard	Occupancy, revenue, open incidents, configurable KPI tiles, AI Copilot panel, predicted wait-times (FR-BI-004).
Incident Console	Active incident queue (incl. Severe Weather per FR-INC-008), SOP checklist execution, response/resolution timers.
Sustainability Dashboard (NEW)	Energy, water, and waste consumption vs. ESG targets, trend reporting (FR-BMS-005).
Video Wall Layout Manager	Camera grouping, layout presets, priority-camera override during incidents.

26.5 Back-Office / Admin Consoles

Screen	Key Content / Function
Ticketing & Pricing Admin	Ticket type setup, pricing rules, promo configuration, refund approval.
Access & Zone Permission Admin	Role-to-zone mapping, blacklist management, override audit log.
HRM & Payroll Console	Employee records, attendance review, leave approval, payroll run.
Finance & ERP Console	GL, AP/AR, budgeting, bank reconciliation, VAT reporting.
Procurement Console	Requisition approval, RFQ management, PO issuance, GRN matching.
Vendor & Sponsor Admin	Contract setup, commission configuration, settlement approval.
Food & Retail Outlet Admin (NEW)	Outlet onboarding, menu/catalog management, stock/prep-time toggles (FR-FNB-007), order queue view.
Maintenance & Work Order Console	Schedule configuration, task assignment, SLA and escalation rules.
System Administration	User & role management, audit logs, integration health monitor, AI model configuration.

26.6 General UI/UX Requirements

- All visitor-facing screens shall support Bengali and English, switchable by the user, with additional languages configurable per FR-CXP-005 for major events.
- All screens shall meet WCAG 2.1 AA (Web Content Accessibility Guidelines) accessibility guidelines at minimum.
- Operator consoles shall support both dark and light mode for extended SOC shift usage.
- Response layouts shall support desktop, tablet, and large-format video-wall resolutions.
- Every admin action that changes access rights, pricing, or financial data shall require an audit-logged confirmation step.

27. API Requirements

27.1 API Design Standards

- All modules shall expose RESTful APIs using JSON payloads over HTTPS (TLS 1.2 or higher).
- APIs shall be versioned (e.g., /api/v1/...) with a documented deprecation policy of at least 12 months' notice.
- Authentication shall use OAuth 2.0 / OpenID Connect with short-lived access tokens and refresh tokens.
- All APIs shall be documented using the OpenAPI 3.x specification and published to a developer portal.
- Event-driven integrations shall be supported via webhooks and/or a message broker (e.g., Kafka/RabbitMQ) for high-volume telemetry (IoT, CCTV analytics).
- All APIs shall implement rate limiting and quota management per consuming application.
- Sensitive endpoints (access control, financial, PII) shall log every call with caller identity, timestamp, and payload hash for audit.

27.2 Representative API Endpoints

Endpoint (illustrative)	Method(s)	Purpose	Module
/tickets	POST	Create a new ticket booking (sport/visitor/premium/membership).	TCK
/tickets/{id}/checkin	POST	Validate a ticket credential at an access point and log entry/exit.	ACC
/visitors	GET/POST	Retrieve or register visitor profile and consent records.	TCK
/access/verify	POST	Real-time identity/credential verification for a given access point and zone.	ACC
/blacklist	GET/POST/DELETE	Manage blacklist entries used by the access-control engine.	ACC
/incidents	POST/GET	Create, retrieve, and update incident records and closure reports (incl. weather-triggered per FR-INC-008).	INC
/incidents/{id}/escalate	POST	Trigger manual or SLA-based escalation of an open incident.	INC

Endpoint (illustrative)	Method(s)	Purpose	Module
/facility/workorders	POST/GET	Create and track maintenance/cleaning work orders.	FAC
/bms/devices/{id}/status	GET	Retrieve live status/telemetry for a BMS-connected device.	BMS
/bms/devices/{id}/command	POST	Send a remote control command to a BMS device (authorized roles only).	BMS
/parking/anpr/event	POST	Receive ANPR read events from parking hardware.	PRK
/hrm/attendance	POST	Submit face/GPS/biometric attendance events.	HRM
/finance/vouchers	POST/GET	Create or retrieve financial voucher/journal entries.	FIN
/vendors/{id}/settlement	GET	Retrieve computed revenue-share settlement for a vendor/period (incl. F&B/retail orders).	VND
/dashboards/kpi	GET	Retrieve current KPI values for a given dashboard/role.	CCC
/ai/copilot/query	POST	Submit a natural-language question and receive a permission-scoped answer.	BI
/food/outlets	GET	Retrieve outlets available for ordering, filtered by the visitor's current zone (NEW).	FNB
/food/orders	POST/GET	Create or retrieve a food/retail order (NEW).	FNB
/food/orders/{id}/status	GET	Retrieve the live status timeline for an order (NEW).	FNB
/webhooks/subscribe	POST	Register a webhook endpoint for a given event type (e.g., incident.created).	ALL

27.3 API Non-Functional Requirements

Attribute	Requirement
Response Time	95th percentile response time ≤ 500 ms for read APIs; ≤ 1.5 s for access-control verification APIs.
Throughput	Access-verification API shall sustain a minimum of 50 requests/second per gate cluster during peak ingress.
Idempotency	All state-changing POST endpoints shall support idempotency keys to safely handle retries.
Error Handling	APIs shall return standardized error envelopes with machine-readable error codes and human-readable messages.
Backward Compatibility	Non-breaking changes only within a major version; breaking changes require a new version.
Sandbox	A sandbox/UAT environment with representative test data shall be available to vendor integration teams and ABG QA.

28. Integration Architecture

ABG-SSCOS shall adopt a hub-and-spoke integration model centered on an Integration Bus / API Gateway. This avoids point-to-point spaghetti integration and allows physical hardware vendors (CCTV, turnstiles, BMS) to be replaced independently of the core platform.

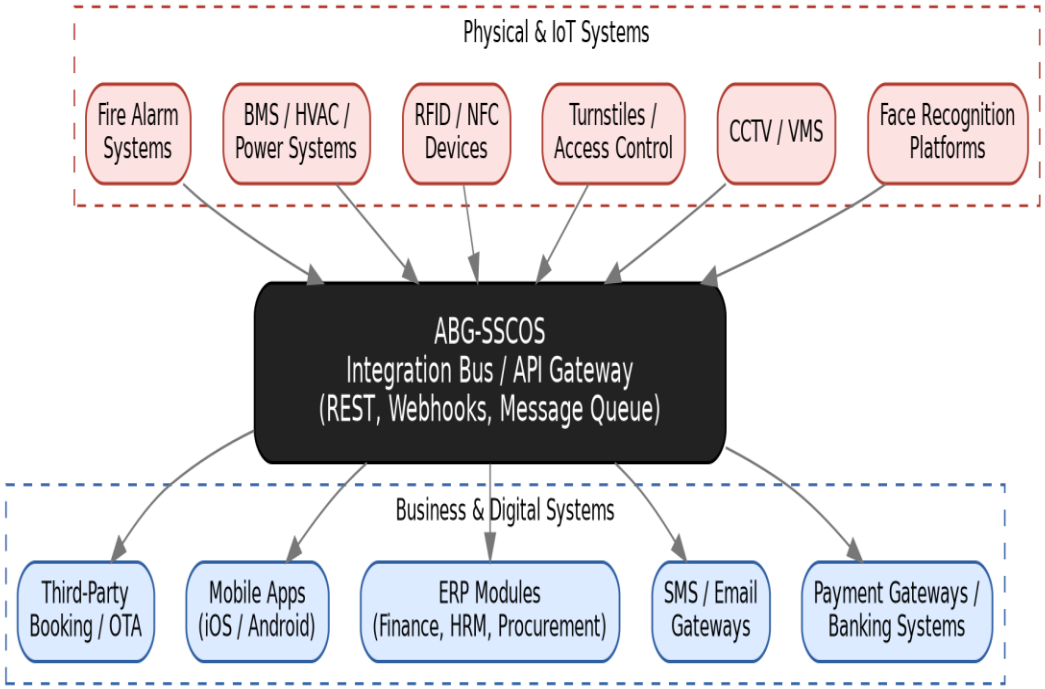


Figure 28.1 — Hub-and-spoke integration architecture via the central Integration Bus / API Gateway.

28.1 Mandatory Integrations

System	Integration Requirement
Face Recognition Platforms	Real-time identity match callback (≤ 1 second) to the Access Control module; enrolment sync API.
CCTV / VMS Systems	RTSP/ONVIF video stream access for AI analytics; event/metadata push to SOC module.
Turnstiles / Access Control Hardware	Bi-directional: credential verification request/response and physical open/deny command.
RFID / NFC Devices	Read-event ingestion and credential lifecycle sync (issue, suspend, revoke), including wearable credentials (FR-TCK-011).

System	Integration Requirement
Payment Gateways / Banking Systems	Payment initiation, callback/webhook confirmation, refund API, settlement file reconciliation.
SMS Gateway	Outbound transactional and alert messaging with delivery-status callback.
Email Gateway	Transactional email (SMTP/API) with bounce/delivery tracking.
ERP Modules (Finance, HRM, Procurement)	Native or API-based integration; no duplicate master data with the core platform.
Mobile Apps (iOS/Android)	Full REST API coverage plus push-notification service (FCM/APNs).
IoT Devices (environmental, occupancy sensors)	MQTT or equivalent lightweight telemetry protocol into the ingestion layer.
Power Systems (substation, generator, UPS)	SCADA/Modbus/BACnet gateway integration into the BMS/Power module.
Fire Alarm Systems	Hard-wired or IP-based alarm signal integration into the Incident Response module, with fail-safe direct alarm sounding independent of software.
Indoor Positioning / BLE Beacon System (NEW)	Real-time visitor zone/location events feeding FR-FNB-002 (food/retail ordering) and reused by FR-DGT-001 (digital twin) — a single positioning integration, not a separate one per module.
Weather Data Feed (NEW)	Live lightning/storm/extreme-heat feed into the Incident Response module to trigger the Severe Weather SOP (FR-INC-008).
Ride-Hailing Platforms (NEW)	API integration (e.g., Pathao, Uber) for coordinated pickup-point suggestion at event end-times (FR-PRK-006).

28.2 Integration Principles

- Life-safety systems (fire alarm, emergency lighting) shall retain independent, hard-wired failure-safe operation regardless of software/network status.
- All integrations shall be resilient to transient network loss with local buffering/replay for at least 30 minutes of data at the edge.
- No single third-party hardware vendor shall require proprietary, closed-protocol integration; ONVIF, BACnet, Modbus, MQTT, and REST/Webhook standards shall be preferred.
- A sandbox integration environment shall be provided per Section 27.3 to validate each integration prior to production cutover.

29. Data Architecture

Data architecture governs how information moves from source systems into a single, trustworthy set of curated data used for dashboards, AI models, and the AI Copilot. The reference flow is shown below; the selected vendor shall propose the specific technology stack (cloud or on-premise) that satisfies the non-functional requirements in Section 3.1.

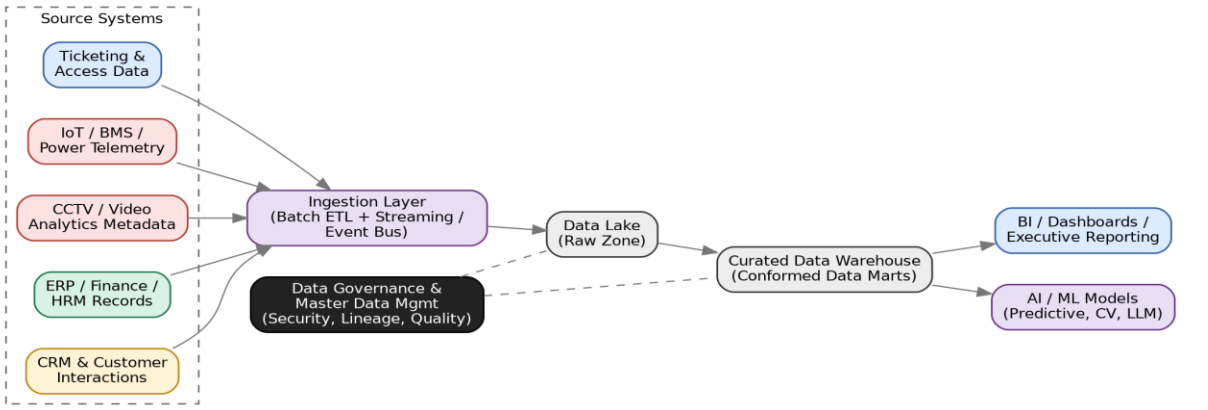


Figure 29.1 — Data architecture from source systems through governance to BI/AI consumption.

29.1 Core Data Domains

Domain	Representative Entities
Identity & Access	Visitor, Member, Employee, Vendor Contact, Credential (incl. Wearable), Blacklist Entry, Zone Permission.
Ticketing & Commerce	Ticket, Booking, Membership, Payment, Refund, Promotion, Invoice.
Facility & Asset	Asset, Maintenance Schedule, Work Order, Inspection Checklist, BMS Device, Sustainability/ESG Reading.
Security & Incident	Camera, Alert, Incident (incl. Severe Weather), SOP Workflow, Response Log, Blacklist Match Log, Lost & Found Item.
Sports & Events	Facility/Court, Slot, Tournament, Fixture, Team, Athlete, Event, Sponsorship Package.
Finance	General Ledger Account, Journal Entry, Cost Center, Budget, Vendor Settlement, Tax Record.
HR	Employee, Attendance Record, Leave Request, Payroll Run, Performance Review.
IoT & Telemetry	Sensor Reading, Power Meter Reading, Environmental Reading, Device Health Event, Indoor Position Event.

Domain	Representative Entities
Food, Beverage & Retail (NEW)	Outlet, Menu/Catalog Item, Order, Order Status Event, Fulfillment Record — reconciled into, not duplicated with, the Vendor Settlement entity above.

29.2 Master Data Management (MDM)

- Visitor/Member identity shall be governed as a single master record across ticketing, access control, CRM, hospitality, and food/retail ordering — no duplicate identity silos.
- Employee identity shall be mastered in HRM and referenced (not duplicated) by attendance, access control, and payroll.
- Asset/Facility records shall be mastered in the Facility Management module and referenced by BMS, Finance (fixed assets), and Procurement.
- Outlet/vendor records shall be mastered in the Vendor & Business Partner module (Section 15) and referenced, not duplicated, by the Food, Beverage & Retail Ordering module (Section 22A).
- A data-quality rule set (mandatory fields, format validation, de-duplication matching) shall be enforced at the point of ingestion.

29.3 Data Governance & Privacy

- PII fields (NID/passport, biometric templates, health data) shall be encrypted at rest with field-level access controls, independent of general database encryption.
- Biometric templates shall be stored as non-reversible mathematical representations, never as raw images, wherever the underlying face-recognition vendor supports it.
- Data retention periods shall be explicitly configurable per domain (e.g., video analytics metadata 90 days per FR-SOC-007, access logs 12 months per FR-ACC-007) with automated purge.
- Data lineage shall be traceable from any BI dashboard figure or AI Copilot answer back to its source system record.
- Consent capture and withdrawal (e.g., for health/performance data under FR-SPT-004, or a minor's wristband pairing under FR-SOC-009) shall be recorded and enforced at the query layer.

29.4 Reporting & Analytics Layer

- A curated data warehouse (or equivalent data-mart layer) shall serve all dashboards to avoid placing analytical load on transactional databases.
- Near-real-time refresh (≤ 5 minutes) shall be supported for Command Center dashboards; batch refresh (daily) is acceptable for deep historical/financial reporting.
- The AI Copilot (FR-BI-002) shall query only the governed, permission-tagged data layer — never raw source tables directly.

30. Key Performance Indicators (KPIs)

The KPIs below shall be available on the relevant dashboards (Section 26.4) and, where noted, referenced in the SLA (Section 32). Targets are initial baselines to be finalized jointly with the selected vendor during design.

30.1 Operational KPIs

Category	KPI	Target (Baseline)	Dashboard
Access	Average gate transaction time (scan to turnstile open)	≤ 2 seconds	Security
Access	False rejection rate (face recognition)	≤ 1%	Security / SOC
Access	False acceptance rate (face recognition)	≤ 0.1%	Security / SOC
Security	Mean time to detect (MTTD) a security incident	≤ 30 seconds	Security
Security	Mean time to first responder dispatch	≤ 3 minutes	Incident Console
Security	Mean time to resolution (by incident severity)	Per SLA tier (Section 32)	Incident Console
Facility	Maintenance task on-time completion rate	≥ 95%	Facility
Facility	Overdue task escalation compliance	100% of overdue tasks escalated within SOP window	Facility
Facility	Unplanned equipment downtime (BMS/Power)	≤ 0.5% of operating hours	Facility
Facility	Utility consumption vs. sustainability target (energy/water/waste) (NEW)	Reduction target set jointly with vendor at design stage	Sustainability

30.2 Commercial KPIs

Category	KPI	Target (Baseline)	Dashboard
Revenue	Daily ticketing revenue reconciliation accuracy	100% match to GL within 24 hours	Revenue / Finance

Category	KPI	Target (Baseline)	Dashboard
Revenue	Online booking conversion rate	≥ 60% of started bookings completed	Revenue
Revenue	Membership renewal rate	≥ 75% annual renewal	Revenue
Revenue	Food/retail order fulfillment accuracy (NEW)	≥ 98% of orders fulfilled without reassignment/refund	Revenue
Vendor	Revenue-share settlement accuracy	100% (zero disputed settlements per quarter target)	Vendor Portal
Vendor	Settlement processing time	≤ 3 business days after period close	Vendor Portal
Sponsorship	Sponsor ROI report delivery time	Within 10 business days of event close	Event/Sponsor

30.3 Customer Experience & AI KPIs

Category	KPI	Target (Baseline)	Dashboard
CX	AI Concierge first-contact resolution rate	≥ 70%	CX Analytics
CX	AI Concierge escalation-to-human handoff time	≤ 60 seconds	CX Analytics
CX	Predicted wait-time accuracy (concessions/restrooms/gates) (NEW)	Within ± 20% of actual observed wait	CX Analytics
AI Copilot	Query answer accuracy (validated sample audit)	≥ 98%	BI / AI Governance
AI Copilot	Average response latency	≤ 5 seconds	BI / AI Governance
Platform	System-wide uptime (transactional)	≥ 99.9% monthly	IT Ops
Platform	System-wide uptime (security/life-safety)	≥ 99.95% monthly	IT Ops

31. Acceptance Criteria

Acceptance shall proceed in staged gates. Each gate requires formal sign-off by the ABG Project Steering Committee before the next stage or payment milestone is released. All acceptance testing shall be conducted against documented test scripts traceable to the Functional Requirement IDs in Sections 4–22A.

31.1 Acceptance Stages

Stage	Scope	Exit Criteria
Design Acceptance	Solution design document, architecture mapping, data model, integration design	100% of Must-priority FRs mapped and approved; no open critical design gaps.
System Integration Testing (SIT)	All modules and third-party integrations (Section 28) tested end-to-end	≥ 98% of test cases pass; zero open Critical/High defects.
User Acceptance Testing (UAT)	Business users execute scenario-based test scripts derived from Use Cases (Section 24)	100% of Must-priority use cases pass; zero open Critical defects; ≤ 5 open Medium defects with agreed remediation dates.
Critical (P1) Defect Closure (NEW)	All defects logged across SIT, UAT, Security Testing, and Load Testing, classified by severity P1–P4	Zero open P1 (Critical) defects of any kind — functional, performance, or security — before go-live approval. P1 is defined as: system-down, data-loss, security-breach, or life-safety-function failure. P1 remediation shall be independently re-verified, not self-certified by the vendor, before this gate is signed off (see also FR SEC-GOV-08 in Section 33.7).
Security & Penetration Testing	Independent third-party penetration test and vulnerability assessment, including targeted P1-severity bug hunting across all modules	Zero unresolved Critical/High vulnerabilities per OWASP/CVSS scoring prior to go-live; any finding scored P1 shall be remediated and re-tested prior to go-live, not deferred to the warranty/hypercare period.
Performance & Load Testing	Simulated peak-event load (Section 3.1 targets)	All latency/throughput targets met at 120% of projected peak load.

Stage	Scope	Exit Criteria
Go-Live Readiness Review	Cutover plan, rollback plan, training completion, support readiness	All prior gates signed off, including the Critical (P1) Defect Closure gate; rollback plan tested; support desk operational.
Warranty / Hypercare Acceptance	Post-go-live stabilization period (minimum 60 days)	SLA targets (Section 32) met for two consecutive full calendar months; zero new P1 defects introduced by warranty-period patches.

31.2 Module-Level Acceptance Criteria (Representative)

Module	Acceptance Criteria
e-Ticketing & Access	100 consecutive test transactions across all ticket types complete with correct pricing, credential issuance, and successful gate entry; blacklist test entries are correctly denied every time.
Security & SOC	Video analytics correctly detect 95% of staged test events (intrusion, object left behind, crowd density breach) within the defined MTTD target.
Incident Response	Each incident type (fire, medical, stampede, security, collision, severe weather) is demonstrated end-to-end from simulated detection to logged closure, with accurate response/resolution time capture.
Facility Management	A full maintenance cycle (schedule → assignment → completion → verification) is demonstrated, plus a deliberately overdue task correctly auto-escalates.
Finance & ERP	Daily ticketing revenue reconciles to the General Ledger with zero variance across a 10-business-day test window.
Food, Beverage & Retail Ordering (NEW)	50 consecutive test orders across at least 3 outlets complete with correct outlet routing by simulated location, correct wallet payment, and revenue that reconciles to the vendor settlement record with zero duplicate or missing entries.
AI Copilot / BI	A defined question set (minimum 30 representative executive questions) is answered with ≥ 98% factual accuracy against source data, with correct permission scoping demonstrated for at least two different roles.
Cybersecurity	MFA, SSO, and encryption controls verified; penetration test closure report accepted with zero unresolved Critical/High findings and zero unresolved P1 items.

32. Service Level Agreement (SLA) Requirements

The selected vendor shall commit to the following SLA framework for the production system, effective from the end of the warranty/hypercare period described in Section 31.1. SLA credits/penalties shall be defined in the commercial contract schedule.

32.1 Availability SLA

System Tier	Uptime Target	Measurement Window	Excludes
Tier 1 — Life-Safety & Security (SOC, Incident Response, Access Control)	99.95%	Monthly	Pre-approved maintenance windows ≤ 4 hrs/month
Tier 2 — Core Transactional (Ticketing, Finance, HRM, Food/Retail Ordering)	99.9%	Monthly	Pre-approved maintenance windows ≤ 4 hrs/month
Tier 3 — Supporting Services (Digital Twin, BI Copilot, Reporting, Sustainability Dashboard)	99.5%	Monthly	Pre-approved maintenance windows ≤ 6 hrs/month

32.2 Incident Severity & Response/Resolution SLA

Severity	Definition	Response Time	Resolution Time
Critical (S1)	Total outage of Tier 1 system, or life-safety function unavailable	≤ 15 minutes, 24/7	≤ 4 hours
High (S2)	Major function of Tier 1/Tier 2 degraded; no workaround	≤ 30 minutes, 24/7	≤ 8 hours
Medium (S3)	Partial degradation with workaround available	≤ 2 hours (business hours)	≤ 2 business days
Low (S4)	Cosmetic or minor issue, no operational impact	≤ 1 business day	≤ 5 business days

32.3 Support & Maintenance

- 24/7/365 Tier-1/Tier-2 support desk for Critical and High severity issues; business-hours support acceptable for Medium/Low.
- A dedicated technical account manager shall be assigned for the first 12 months post go-live.
- Quarterly security patching cadence at minimum, with emergency patching for actively exploited vulnerabilities within 72 hours of vendor patch release.
- Monthly SLA performance reports shall be submitted to the ABG Project Steering Committee, including root-cause analysis for any S1/S2 incident.
- Backup verification (restore test) shall be performed and evidenced at least quarterly.

32.4 Data Protection & Continuity SLA

System Tier	**RPO**	**RTO**
Tier 1 — Life-Safety & Security	≤ 1 hour	≤ 1 hour
Tier 2 — Core Transactional	≤ 15 minutes	≤ 4 hours
Tier 3 — Supporting Services	≤ 24 hours	≤ 24 hours

33. Cybersecurity Requirements

Given the platform's role as a security, financial, and life-safety system, cybersecurity requirements are treated as first-class, non-negotiable requirements rather than a generic checklist.

33.1 Compliance Standards

Standard	Applicability
ISO/IEC 27001	Information Security Management System (ISMS) covering the vendor's development and hosting operations.
ISO/IEC 27701	Privacy Information Management, applicable given the volume of visitor/employee PII and biometric data processed.
SOC 2 Type II	Required for any cloud-hosted component, covering security, availability, and confidentiality trust principles.
OWASP Top 10 / ASVS	Secure application development lifecycle for all web/mobile/API components.
PCI DSS	Required if the vendor's platform directly touches card payment data rather than fully tokenizing via the payment gateway (applies equally to the wallet used for ticketing and food/retail ordering).
Bangladesh Data Protection Act (as applicable/enacted) & sector regulations	Data residency and citizen-data handling obligations.

33.2 Identity & Access Management

- Multi-Factor Authentication (MFA) shall be enforced for all administrative, financial, and SOC operator accounts.
- Single Sign-On (SSO) via SAML 2.0 / OIDC shall be supported for enterprise identity provider integration (e.g., Azure AD).
- Role-Based Access Control (RBAC) shall be enforced at both the UI and API layer, with least-privilege default profiles.
- Privileged account activity (super-admin, database admin) shall be recorded and subject to periodic access review (minimum quarterly).
- Session timeout, password complexity, and account lockout policies shall be configurable to ABG's security policy.

33.3 Data Protection

- Encryption at rest (AES-256 or equivalent) for all databases, file stores, and backups.

- Encryption in transit (TLS 1.2+) for all internal and external network communication, including IoT/edge device links where technically feasible.
- Field-level encryption and access segregation for NID/passport numbers, biometric templates, and health data, per Section 29.3.
- Key management via a dedicated Hardware Security Module (HSM) or cloud KMS, with documented key-rotation procedures.

33.4 Application & Infrastructure Security

- Secure Software Development Lifecycle (SSDLC) including static (SAST) and dynamic (DAST) application security testing prior to each major release.
- Dependency/software composition analysis (SCA) to detect vulnerable third-party libraries.
- Network segmentation separating Tier 1 life-safety/security systems, Tier 2 transactional systems, IoT/OT networks, and public-facing DMZ components.
- Web Application Firewall (WAF) and DDoS mitigation for all public-facing endpoints.
- Endpoint protection and hardening baselines for all operator workstations in the Command Center and SOC.

33.5 Monitoring, Detection & Response

- A Security Information and Event Management (SIEM) platform shall aggregate logs across application, infrastructure, and network layers.
- SOC monitoring (cyber, distinct from the physical Security Operations Center in Section 6) shall provide 24/7 threat monitoring with defined escalation procedures.
- AI-based threat analytics shall be used to detect anomalous access patterns, credential stuffing, and insider-threat indicators.
- A documented Incident Response Plan (cyber) shall define roles, communication protocols, and regulator/customer notification timelines consistent with applicable law.
- Immutable audit logs shall be retained for a minimum of 12 months (or per Section 29.3 domain-specific retention) and protected from tampering.

33.6 Third-Party & Supply Chain Security

- All third-party hardware/software integrations (Section 28) shall undergo a security review prior to production connection.
- Vendor personnel with access to production systems shall be subject to background checks and signed confidentiality/NDA agreements.
- The vendor shall disclose its own subcontractors and hosting providers, including data-center locations, for ABG's data-residency assessment.

33.7 Governance & Assurance

- Independent penetration testing shall be conducted at least annually and after any major release, per Section 31.1.
- The vendor shall provide a right-to-audit clause allowing ABG or its appointed auditor to review security controls.
- A responsible-disclosure / bug-bounty channel shall be established for reporting vulnerabilities.
- Security awareness training shall be provided to ABG operational staff as part of the knowledge-transfer deliverables.

Important Note:

SEC-GOV-08 (NEW): The vendor shall maintain a P1 (Critical) defect and vulnerability register spanning every test cycle — SIT, UAT, penetration testing, and load testing. No release, including go-live, shall proceed with an open P1 item. P1 remediation shall be independently re-verified (not self-certified by the vendor) before closure sign-off by the ABG Project Steering Committee, consistent with the Critical (P1) Defect Closure gate in Section 31.1.

34. Vendor Evaluation Framework

This section defines how vendor proposals will be scored. Vendors are strongly encouraged to structure their proposal response in the same order as this document to simplify evaluation.

Evaluation Category	Weight	What Is Assessed
Functional Compliance (Sections 4–22A)	25%	Coverage and compliance status against every FR-ID, including configuration vs. customization effort.
Technical Architecture & Integration (Sections 3, 27–29)	20%	Architecture soundness, API maturity, integration approach, data governance capability.
Security & Compliance (Section 33)	15%	Certifications held, security architecture, audit history, breach disclosure record, and P1-defect handling track record.
AI & Automation Capability (Sections 5–7, 21–22)	15%	Maturity of face recognition, video analytics, predictive maintenance, and AI Copilot capability, with reference deployments.
Implementation Plan, SLA & Support (Sections 31–32)	10%	Realism of timeline, staffing plan, SLA commitments, support model.
Commercial Proposal	10%	Total cost of ownership over 5 years, licensing model, payment milestones.
Vendor Track Record & References	5%	Comparable smart-city/stadium/sports-complex deployments, client references.

34.1 Proposal Submission Requirements

- Point-by-point compliance matrix against every FR-ID, US-ID, and UC-ID in this document (including the new FR-TCK-011, FR-SOC-008/009, FR-INC-008, FR-BMS-005, FR-PRK-006, FR-CXP-005, FR-BI-004, and the FR-FNB series).
- Proposed solution architecture diagram mapped to the six-layer reference architecture (Section 3).
- Sample API documentation (OpenAPI spec) for at least three representative modules.
- Security certifications (copies) and most recent penetration test executive summary (redacted as needed).
- Implementation project plan with phasing, milestones, and resourcing.
- Minimum three reference clients with comparable scale, contactable for due diligence.
- Commercial proposal broken down by module/layer, with clear distinction between one-time and recurring costs.

34.2 Recommended Procurement Timeline

Milestone	Indicative Timing
RFP Issuance	Week 0
Vendor Q&A Period	Week 1–2
Proposal Submission Deadline	Week 5
Technical Evaluation & Shortlisting	Week 6–7
Vendor Demonstrations / Proof of Concept	Week 8–9
Commercial Negotiation & Contracting	Week 10–12
Design Acceptance Kickoff	Week 13

Appendix A — Requirement Traceability Index

A full traceability matrix (Functional Requirement → User Story → Use Case → Test Script → Acceptance Criterion) shall be maintained as a living spreadsheet artifact alongside this document and issued to the selected vendor at Design Acceptance kickoff. The ID conventions used throughout this document (FR-xxx-###, US-##, UC-xxx-##) are structured to support direct import into the vendor's and ABG's requirements management tooling (e.g., Jira, Azure DevOps, or a dedicated RTM spreadsheet).

v2.0 ID register check: all new IDs introduced in this revision — FR-TCK-011; FR-SOC-008/009; FR-INC-008 (with FR-INC-001 edited in place); FR-BMS-005; FR-PRK-006; FR-CXP-005; FR-BI-004; FR-FNB-001 through FR-FNB-011; US-17; UC-FNB-01; and SEC-GOV-08 — were checked against the full existing ID sequence in Sections 4–22 and 33 before assignment. No ID collides with an existing one, and no new ID restates a requirement already owned by another section; each new item either extends an existing FR by explicit cross-reference or introduces a genuinely new capability.

Appendix B — Glossary

Term	Definition
ANPR	Automatic Number Plate Recognition
BMS	Building Management System
ESG	Environmental, Social and Governance — used here for the sustainability reporting dashboard (FR-BMS-005)
FNB	Food, Beverage & Retail Ordering module prefix used for requirement IDs in Section 22A
MTTD / MTTR	Mean Time to Detect / Mean Time to Resolve
P1	Priority/Severity 1 (Critical) — a defect or vulnerability representing system-down, data-loss, security-breach, or life-safety-function failure; see Section 31.1
RPO / RTO	Recovery Point Objective / Recovery Time Objective
SOC	Security Operations Center (physical/cyber, as noted per section)
SOP	Standard Operating Procedure
SIEM	Security Information and Event Management
RTM	Requirement Traceability Matrix